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Media Project

Evaluation of Haptics with Auditory Cues in Virtual Reality

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1 Introduction

The most important aspect of virtual reality (VR) is the degree by which a user can feel and experience virtual space as though it is reality. Multimodal interaction can enhance user experience by using multiple modalities such as haptic and auditory feedback. Haptic and auditory feedback in addition to visual representation can create a more realistic and engaging experience for the users. Haptic Feedback addresses the human's sense of touch and feel of the objects by simulating force and vibrotactile feedback. To incorporate haptics in Virtual Environments, haptic devices such as haptic gloves can produce forces and vibration, are used to enhance the experience, and have a more realistic perception of users of the object they are interacting with.

Gardening is a hobby of a lot of people, but many kinds of research have proven that engagement with natural environments and representations of nature has contributed to improving mood states and reducing cognitive decline for people[1]. This is what we focused on as well as to develop a VR environment in which people can do gardening and the more fun part is that there will be not only visual engagement but sensory engagement as well. The high-end implementation of a gardening environment in Virtual Reality using Haptics with Auditory cues will contribute to a fun and lively experience for people in the modern era.

For the virtual reality experience, the demand remains for an interactive system with an even greater immersion and sense of reality through feedback that can satisfy the five senses, including visual, auditory, and tactile, to enhance the user's sense of presence. Immersion allows users to experience where they are and what they are doing as if it's a real experience[2]. In addition, immersion is defined as the extent of engagement a user experiences[3].

The concept of presence refers to a phenomenon where users act and feel as if they are "being there" in the virtual world. Consequently, immersion is a term used for describing the technology that can give rise to presence[4]. Perceptual realism is how a realistic user perceives the objects which he is looking at and interacting with. Without haptic feedback, interactions with objects in the VR environment can lead to a gap between the actual and virtual realities; therefore, feedback from such interactions with objects through a haptic system is crucial in order to accurately express the interaction between the virtual object and reality[5]. This is an essential step in enhancing the sense of presence, realism, immersion, and interaction with objects in VR applications.

2 Related Work

A Lot of work has been done for haptic feedback in the past. In 1994, Richard et al. researched and examined the behavior of haptic feedback with Artificial sound on task performance while grabbing and moving the virtual ball, exploring that haptic feedback is better for hard balls whereas it sounds better for softballs[6]. That shows that the research on haptic feedback has been going on since the 90's.

A study was conducted at Nuremberg Institute of Technology in Germany with the title "Evaluation of different types of haptic feedback influencing the task-based presence and performance in virtual reality" in which they examine the influence of different types of haptic feedback on presence and performance regarding manual tasks in virtual reality[7]. Their study includes 14 subjects who performed throwing, stacking and object identification tasks in virtual reality with visual, vibrotactile and force feedback. They used HTC Vive and Sense Gloves with the Leap Motion Controller for hand tracking and force feedback. The data collected by them was from questionnaires and the task performance was measured by considering the total task execution time. In the first task of throwing, their results showed that haptic feedback did not increase the hit accuracy or rate significantly, but significantly lowered the time needed to complete the task. In the second task of stacking, again there was no significant task performance related improvement occurred with vibrotactile compared to visual feedback. In the last task of object identification, the result showed that vibrotactile feedback did enhance the object detection rate but also took more time because sensory information takes a longer processing time.

Another research study was conducted in RWTH Aachen University in Germany with the title of "Comparing Auditory and Haptic Feedback for a Virtual Drilling Task"[8]. The goal of the study was to examine how auditory feedback can be used to substitute haptic feedback when interacting with a vibrating tool. They did a study by conducting a drilling task on 8 participants and examined the combination of Auditory feedback, Vibrotactile feedback, and the Surface Force feedback on User Performance. The main measurements of the task were the amount of removed target material and error measured as the amount of erroneously removed surface material. The study result showed that both Auditory and Vibrotactile Feedback has a similar influence on User Experience because both led to similar changes whereas the surface force feedback just provides the information about the contact with the surface but not about material type. Vibrotactile and Auditory Feedback have similar improvement in performance whereas combining both does not have any significant improvement. The comparison shows that Auditory

and Vibrotactile feedback can be combined with the other feedback type, but surface force feedback cannot be combined with other feedback types and can stand alone.

A study was conducted at St George's University Hospital in London with the title “Impact of haptic feedback on surgical training outcomes: A Randomized Controlled Trial of haptic versus non-haptic immersive virtual reality training “where they evaluated and compared the performance for an orthopedic surgical task with haptic feedback in a VR-based Simulation with the one with non-haptics[9]. The study showed that the evaluation with haptic feedback shows better performance. They performed a bone drilling task, and the purpose was to teach the process with the help of VR Simulation and haptic feedback, and they found out that the group with haptic feedback take much more drill time than the non-haptic group and the safe drills of <5mm plunge gap in at least 2 out of 3 attempts.

3 Implementation

3.1 Hardware

- HTC Vive Pro Eye

Fig. 1 HTC Vive Pro Eye

- Sense Gloves (DK1)

Fig. 2 Sense Gloves DK1

- HTC Vive Trackers (3.0)

Fig. 3 HTC Vive Trackers 3.0

- Sense Gloves Wireless Kit

Fig. 6 Sense Gloves Wireless Kit

3.2 Software

- Unity 3D v2020.3.19f1
- Sense Gloves SDK v2.5.0
- Steam VR v1.6.10
- Blender v3.6
- Visual Studio Community v2019
- Anaconda v2023.03-1
- Jupyter Notebook v6.5.2

3.3 Programming Languages

- C#
- Python

3.4 Implementation

3.4.1 Environment Creation

The environment was created inside unity manually without any auto terrain tool by using different assets from the Unity Asset Store and some of the assets were modified later according to the requirement of the environment using Blender. Gardening tools such as

shovel, pitchfork and rake, a greenhouse with plants inside, tree, different plants and vegetables, storage baskets, seed bags, fertilizer bags, water tank, watering can, pots, plantation areas, raised beds, bench, task boards, water barrels, wheelbarrow, table, fences, cat and honeybee were added to give it a complete garden look. The skybox was meticulously customized to showcase the weather condition which was mostly cloudy with the sun coming out of the clouds. The customization of the sky gives a more realistic and natural look to the environment. To enhance the ambiance, we implemented the rain particle system which simulates rainfall. VR Interactive buttons were also designed and placed in the environment such as on the task's boards and on the water tank.

Fig. 7 Task Board in Environment

Fig. 8 Overall Environment

3.4.2 Haptic Feedback

To implement the haptic feedback, algorithms related to sense gloves were used. The algorithms send the vibrotactile feedback according to the different intensity level and also delivers the force feedback according to the different magnitudes and distance such as when grabbing a shovel in a haptic environment will provide a certain vibration to your hand and picking the tomatoes will provide force feedback in the haptic environment or feeling the sting of the honeybee on your finger by feeling the vibration on that particular finger. You wear the Sense Gloves DK1 on the back of your hand onto which are attached the five moving fingers, five rigid plastic sticks are presented as five fingers and there is a black plastic connected that can rotate pivoting around the top of the blue stick. A string is attached between the blue and the black plastic which is responsible for the force feedback, basically you control the magnitude and distance so the string gets tightened, and you can feel the force feedback. We consider the maximum force and the maximum force distance for the force feedback, the maximum force defined as the maximum brake force that the material provides when the finger is at maximum force distance inside the collider. The force value is the range between 0 to 1 whereas the maximum force distance is the distance in meters before the maximum force is reached. There is a motor attached to each finger which is responsible for the vibrotactile feedback, with the change of intensity we can feel and differentiate between different vibrotactile feedbacks. Basically, you have to define which hand you want to send the vibrotactile feedback to, it can be both hands or one of the hands depending upon the situation and also you define an array which takes the number of fingers, and you can specify the fingers on which you want a vibrotactile feedback. For vibrotactile feedback we have considered a variable magnitude which is a range between 0 to 100 and the corresponding finger and duration of the vibrotactile feedback.

Objects	**Intensity**
Shovel	60
Pitchfork	40
Rake	20
Seed Bags	30
Watering Can	40
Tomatoes	55
Grapes	40
Rain	20
Cat	20
Honeybee	50

Table 1. Intensity level during vibrotactile feedback

Objects	Max Force	Max Force Distance
Pot	1	0.04
Pumpkin	1	0.03
Tomatoes	0.5	0.02
Grapes	0.25	0.03

Table 2. Maximum force and maximum force distance (in meters) during force feedback

3.4.3 Arm Swinging Locomotion:

To ease the movement within the environment, arm swinging locomotion was implemented using which you can move within the environment by just swinging your arms in real-time standing at the same spot and your position will change frame by frame, taking the default gravity factor into account. The arm-swinging locomotion depends on the motion of the hands, when the user swings their arms as if walking or running, the system detects the arms swing and translates it into the virtual movement. The speed of the movement depends on the intensity of the arms swing, swinging the arms faster can make users move faster and slower swings can slow down the movement. It takes the direction from the user's head movement and wherever the user is looking, it creates the mesh in that direction and moves the user forward in that direction. It is a continuous movement and gives the user a more natural, realistic and immersive feeling compared to other locomotion techniques in Virtual Reality. It provides less motion sickness as compared to other movement techniques such as teleportation.

3.4.4 AI Based NavMesh:

For the movement of the characters such as cat in our scenario, we used Artificial Intelligence-based NavMesh which enables intelligent pathfinding and movement of the characters. It consists of interconnected polygons which determine the path for the character. It involves defining walkable areas and creating a navigation mesh which can allow the cat to navigate along predefined paths in the virtual garden.

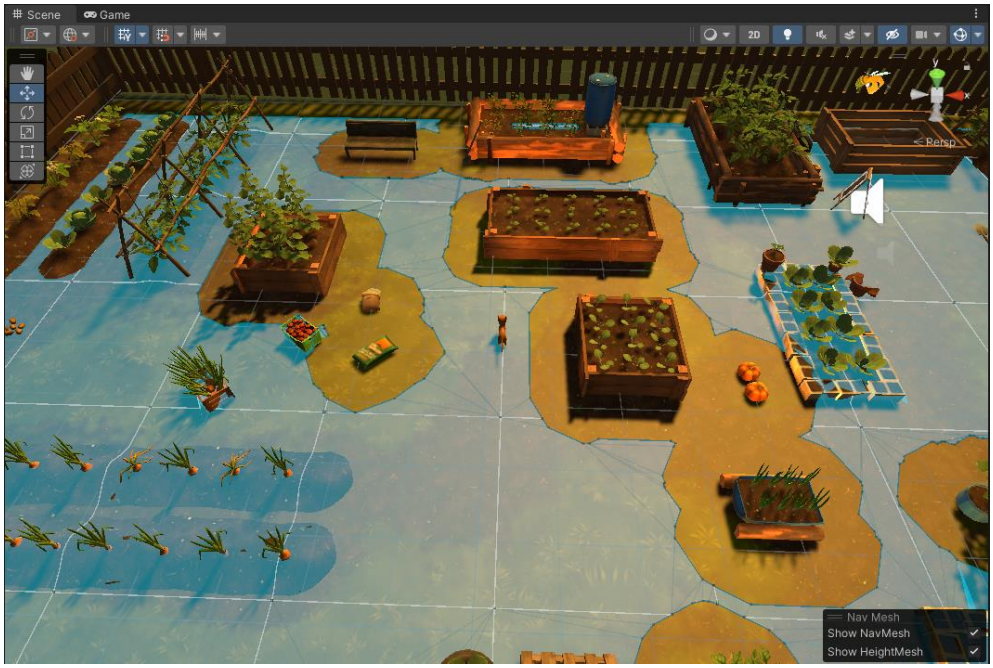

Fig. 7 AI based NavMesh

3.4.5 *Particle System:*

To create the visual effects in our environment such as rain effect, water effect while putting water from tank to watering can or from watering can to the plantation area, we used particle system algorithms. These algorithms generate and control the behavior of individual particles to create realistic and dynamic effects in the virtual environment.

3.4.6 *Auditory Cues:*

The addition of sound gives a more realistic and engaging experience to the virtual environment such as the sound of rain, sound of honeybee, breaking sound of the pot and ambient bird sounds contributes to the overall immersion of the environment. The auditory cues when combined with the haptic feedback enhance the perception of touch and the sense of presence within the virtual environment. In the project we combine the sound effect of rain with vibrotactile feedback and generate random rain phenomena so whenever it rains the user will feel vibrotactile feedback on his fingers and give him the feel that the rain droplets are coming down and touching his fingers. The breaking sound where the participant breaks the pot is combined with the force feedback where he has to apply some force in order to break the pot gives a natural feeling and engages him more into the virtual environment, the auditory cues when the honeybee is near you gives you a sensation that honeybee is approaching you or nearby you which engages your senses.

3.4.7 Finger-Specific Feedback:

In addition to hand feedback, finger-specific feedback was also implemented to have vibrations in those specific fingers only which will touch the objects. Specifically for the tasks such as picking up grapes from the plant, the participant only needs his two fingers to pick the grapes due to their small in size compared to tomatoes, so we specifically sent the vibrotactile and force feedback to only those two fingers in order to achieve a more precise and localized haptic experience, mimicking the real life sensation of using fingers to pick the objects.

3.4.8 Animations:

Animations are a very important part of making the virtual environment immersive and look realistic, also combining the animations with haptic feedback enhances the overall user experience. We used different animations for different purposes such as the bee flying animation and cat animations which were combined with haptic feedback, and some navigation animations were also used to help the participant in finding the next task board in the environment. The walking and sitting animation of the cat with the combination of vibrotactile feedback where you can touch the cat to show your affection and receive vibrotactile feedback. Moreover, the animations of the cat's sitting and vocalizing "meow" responses enhanced the overall sense of presence and engagement, making the environment more alive and responsive to the user's action.

3.4.9 Plant Growth:

Sow the seeds in the soil, water them and they will grow into a proper plant, that is what was replicated in implementation as well. Specific to what seeds were sowed in soil, on watering the seeds, the seeds will grow into a specific proper plant of which the seeds were sowed. Plant growth will be in stages, considering specified max growth stages, with some time gap of specified intervals and hence users can see plant growing in different stages.

3.4.10 Way Points:

Way points were implemented for the movement of the honeybee. Way points are predetermined locations which serve as key navigation points for the honeybee. Instead of following a fixed pattern and to make the honeybee more dynamic, the honeybee chooses its next way point randomly from the available way points. The user gets vibrotactile feedback on his finger when the honeybee stings his finger. and to make it more realistic we have added the 3D sound of the bee to make the user aware that the bee is nearby him. To avoid the disturbance, we have manually set the minimum and maximum distance for the 3D sound so it can only be heard when the honeybee is near to the user

and the user cannot hear the honeybee sound when the honeybee is roaming somewhere else in the virtual garden.

3.4.11 Study Specific Task Creation

We created two environments to conduct the study, one with haptic feedback and the other without haptic feedback. Considering our experimental design, 4 tasks were implemented in both environments. In each task, we introduced a task board which has all the steps written on it and these steps must be completed by the participant. A Virtual Button was designed on the board which needs to be pressed to activate the timer. The timer only gets visible when the participant presses the timer button and stays visible throughout the task. After the completion of all the steps written on the board, a green tick will display on the board indicating the successful completion of the task. After completing all the steps by the participant, the timer gets stopped automatically on its own and displays the time which has been taken by the participant. In task 1, the participant has to break the pot and deform the pumpkin. To break the pot, the participant experienced the force feedback whereas to deform the pumpkin, the participant experienced the combination of force feedback with vibrotactile feedback. In task 2, the participant has to grab shovel, pitchfork and rake one by one and put them into the storage area. On grabbing the shovel, pitchfork and rake the participant experienced the vibrotactile feedback with different intensity, shovel with the highest intensity of 60 whereas the pitchfork intensity was 40 and rake was having 20 intensity level. In task 3, the participant has to grab the seed bags, and put the seeds in the plantation area. After putting the seeds, he needs to go to the water tank and fill the watering can by pressing the VR button which is there on the water tank. After filling the water tank he has to go back to the plantation area and water the seeds. On grabbing the seeds bag and watering can, the participant will receive vibrotactile feedback whereas the particle system will get active when he will press the button to fill the water into the watering can and also when he is watering the plantation area from the watering can. Particle system was combined with auditory cues to give a more realistic feel. In task 4, the participants have to pick 2 tomatoes from the tomato plant and put them into the basket. On grabbing the tomatoes, he will receive force feedback with the combination of vibrotactile feedback. Then he has to pick 2 grapes and put them in the basket with his two fingers so he can experience the finger specific feedback as well. In case of a nonhaptic environment, the participant has to go through the same 4 tasks but there will be no force feedback, finger specific feedback or vibrotactile feedback in the non-haptic environment and he will only have the visual and auditory cues.

The study time for each participant was 60 minutes and due to the limited time, we did not introduce cat, rainfall, and honeybee in the study. Another reason for not including them was complexity, while having a vibrotactile feedback on the rainfall event and participant picking the shovel at the same time will make him confused and he cannot identify both vibrations at the same time. As rainfall was randomly generated and honeybee was also randomly comes and sting the participant so we removed it from the study as it can also create confusion if the participant is grabbing something and feeling vibrotactile feedback and honey bee stings him the same time, than it will be very hard for him to differentiate and identify two vibrations at the same time. So, by considering the complexity and the time constraint, we did not include them in the study and put them separate from it.

4 Evaluation

4.1 Research Questions

To evaluate the impact of haptic feedback on the user's ability to interact with the environment and to assess the user's perception about the virtual gardening environment we have investigated it for 5 research questions by conducting a study. These research questions majorly focus on investigating the impact of haptic feedback on 5 metrics, spatial presence, perceptual realism, mental immersion, user experience, and interaction of the user based on task performance time in the environment. Research questions with the proposed hypothesis regarding the study conducted are as follows:

1. Question: To what extent users experience a sense of presence there inside the environment with haptic feedback?

Hypothesis:

With haptic feedback users will report higher levels of spatial presence within the environment compared to when there will be no haptic feedback.

2. Question: What impact does the haptic feedback have on the user's perception of objects realism in the virtual gardening experience?

Hypothesis:

Users who experience the haptic feedback will evoke higher perceptual realism, compared to the absence of it.

3. Question: What does the effect of haptic feedback have on the user's feeling of being mentally immersed in the environment?

Hypothesis:

Users who experience haptic feedback will feel more mentally immersed in the environment compared to when there will be no haptic feedback.

4. Question: What impression does presence and absence of haptic feedback have on user experience?

Hypothesis:

Users will have better user experience with haptic feedback as compared to absence of it.

5. Question: What impact does the haptic feedback have on the user's ability to interact with virtual gardening objects and perform tasks effectively?

Hypothesis:

Users who experience the haptic feedback will lead to shorter task performance time, compared to when they lack haptic feedback.

4.2 Participants

A total of 19 participants were recruited for the study with no restriction on age. Among 19 participants 10 were male and 9 were female. The result for 1 male participant was discarded due to hardware issues. An equal number of male and female participants were considered for the study (mean age of 26.94 years with a SD of 2.38). Participants who needed vision correction had to wear glasses under the HMD. Among participants whose data was considered six participants had no experience of VR and thirteen had experience with VR.

4.3 Experimental Design

4.3.1 Objective

The objective of this study was to investigate the impact of haptic feedback on 5 metrics, spatial presence, perceptual realism, mental immersion, user experience and interaction capability, in a virtual gardening environment.

4.3.2 Independent Variables

The independent variable in the study was haptic feedback. There were two conditions:

- Condition 1: Gardening tasks with haptic feedback.
- Condition 2: Gardening tasks without haptic feedback.

4.3.3 Metrics

A total of 5 metrics which were dependent variables in the experiment were considered. Those dependent variables are:

- Perceptual Realism: Participants subjective ratings (Likert scale) of their perception of objects realism.
- Task Performance: Task completion time for every specific task.
- Spatial Presence: Participants subjective ratings of their sense of presence within the virtual gardening environment.
- Mental Immersion: Participants subjective ratings of their feeling of being immersed in the virtual gardening environment.
- User Experience: Participants subjective ratings of their impression and overall experience with the gardening virtual environment.

4.3.4 Questionnaire Design

A proper questionnaire was designed to gather participant data, age, gender, and whether he already had any VR experience using any HMD. For every metric to measure, questionnaire questions were taken from already designed standard questionnaires and incorporated together to perfectly design the questionnaire as per our experimental need. For investigating spatial presence within the environment, the self-location questions were taken from the Social Presence Experience Scale (SPES) questionnaire with Likert scale values 1 to 7. For investigating perceptual realism and mental immersion the questions were taken from the Telepresence Inventory questionnaire (TPI), with Likert scale values 1 to 7. To assess user experience, Attrakdiff questionnaire's short version was used with Likert scale values from -3 to +3.

4.3.5 Experimental Setup

The experimental setup involved a virtual reality (VR) system which provides the gardening environment. Participants had to wear VR headset - HTC Vive and Sense Gloves on their hands (Fig. 8), along with Sense Gloves wireless kits to interact with the virtual objects and experience haptic sensations. Auditory cues were incorporated into the virtual environment to accompany haptic feedback when applicable. Two environments were used for the study, one with haptic feedback and the other without haptic feedback. The counterbalancing technique was considered as well for experiment execution. Counterbalancing is a technique used to deal with order effects when using a repeated measures design. With counterbalancing, the participants sample is divided in half, with one completing the two conditions in one order and the other half completing the conditions in reverse order. So, considering this one set of 9 participants experienced a haptics environment first and then non-haptics and the other 9 participants experienced in the reverse order.

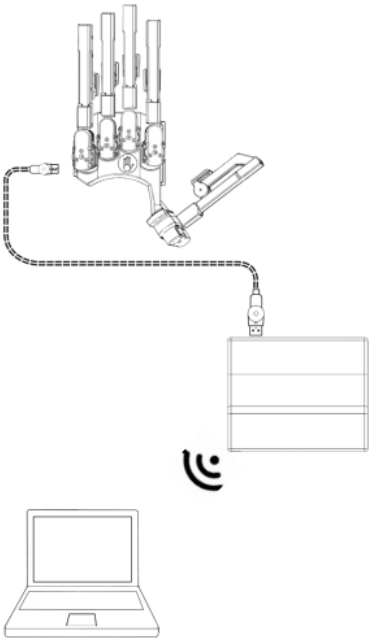

Fig. 8 Experimental Setup of HTC Vive with Sense Gloves

4.3.6 Experimental Procedure

Firstly, to every participant a brief overview was given in context of the purpose of the study and what hardware devices were used in the experiment. Then he was asked to sign a consent form for participating and contributing to the experimental study. After that to guide about complete experimental flow, were presented with a proper experiment flow guide document, in which it was explained what steps will be followed for experiment execution. The hand calibration was done before proceeding with the actual experimental task execution. They were guided about how they move within the environment by swinging their arms so that with locomotion they can move around freely and explore and interact with the environment. Then they were trained using a training task in which participants got familiar with object interaction, such as grabbing, throwing, and breaking by including some force feedback.

After this, they were presented with an actual gardening environment in which they performed 4 tasks and experienced haptic feedback. Task boards were used for each task, there were 4 boards in total and each of them provided the complete steps for each task. Animations were also used to help the participant in locating different areas and task boards. Participants were navigated within the environment by an arrow animation which was displayed on the currently active task. All task times were monitored with a task timer which started by pressing a button attached to the instruction

board. In task 1, the participant starts the task by pressing the timer button on the board to start the timer, after that participant has to grab a pot and then break it by applying force. On applying force, the participant received the force feedback with the combination of auditory cues when the pot breaks in the form of a breaking sound effect which was synchronized with the virtual feedback providing a realistic and engaging experience. The next step was to grab a pumpkin and deform it, on grabbing the pumpkin the participant received the vibrotactile feedback and to deform the pumpkin a sufficient amount of force was required which instantiate the force feedback with vibrotactile and visual feedback, the more force participant applied, the more deformation took place. The last step was to put the pumpkin in a nearby basket. On completing the task, the timer stopped, and a green arrow sign appeared on the bottom of the board indicating that the participant had completed the task successfully and enhancing it by introducing the auditory cues in the form of an audio sound saying, “task completed, go to the next task”.

To help users to find the next task, which was task 2, arrow animations helped the participant in finding the next task board. After reading the task steps, the participant pressed the timer button to start the timer. Task 2 was based on vibrotactile feedback with different intensities on grabbing different objects such as shovel, pitchfork, and rake which the participant has to put in a storage area. The timer stops and a green arrow sign pops up on the bottom of the task board indicating that the participant has successfully completed the task with an audio sound of “task completed, go to the next task”.

Task 3 was the lengthy one in all 4 tasks, here the user felt the vibrotactile feedback with both auditory and visual cues. In task 3, participants have to grow a plant on their own. After carefully reading the steps from the board, the participant pressed the timer button to start the timer. There were 4 seed bags on the table which were of onion, cucumber, tomato, and carrot seeds. Participant has to grab any of the seed bags which were on the table and had to put the seeds in a plantation area which was indicated by the arrow animation, after that he has to move towards the water tank and filled the water in a watering can by pressing the button of the water tank. There he received the auditory cue in the form of a water dripping sound from the tank to the can. Then he took the watering can to the plantation area and watered the seed. The plant was grown in 3 stages, and the seeds which he put in at the start determined which plant would be grown there. Multiple plants were also grown for some participants who took 2 seed bags and put both in the plantation area. The timer stopped and the green tick sign on the bottom of the board displayed an audio sound of “task completed, go to the next task”. The last task was focusing on force feedback, vibrotactile feedback, and finger-specific vibrotactile feedback where the participant had to pick 2 tomatoes from the tomato plant to the basket with his hand and then 2 grapes from the grape plant with his fingers receiving the vibrotactile feedback to only the fingers which were in contact with the grapes. Before starting the task,

participants read the steps carefully and pressed the timer button. After completing the task, a green arrow sign indicates that the participant has successfully completed the task.

Time was noted for all four tasks performed to analyze the effect of haptic feedback on user interaction and task performance in terms of time. After completing all tasks with haptic feedback, the participant filled out a questionnaire considering his experience. After filling out the questionnaire, the participant was brought back to the virtual environment which looks almost same and this time he again performed the same 4 tasks which were performed earlier but without haptic feedback. Again, the participant filled out the same questionnaire considering his non-haptics experience. All the data gathered from questionnaires and tasks time was stored in two CSV files, named “Haptics” and “Non-Haptics” for further analyzing it.

4.3.7 Analysis Approach

To test our hypothesis for the proposed research questions, qualitative data was gathered in two CSV files “haptics” and “non-haptics” for each environment, from questionnaire ratings provided by participants. Data was analyzed using Python programming language in Jupyter Notebook. Step-by-step analysis was performed in which firstly type of variables was determined if they were categorical, continuous, or ordinal variables.

Age was a continuous variable, gender and HMD usage were categorical variables as these have qualitative characteristics or attributes that belong to distinct categories. Among the rest of the variables, the Likert scale questions variables were considered ordinal variables. Ordinal variables are often used to measure subjective opinions or attitudes, and they are frequently analyzed using non-parametric statistical tests that do not assume a normal distribution. First, all rating data was converted to numeric form, the overall rating for each participant by averaging across all columns was calculated. After that from the Python “Scipy” library Wilcoxon signed-rank test was used for the analysis of data and computation of the p-value.

Since these are ordinal variables so no normality test was performed to check the normality of the variables. Also, no outliers were checked, since outliers are data points that significantly deviate from the majority of the data. In the case of Likert scale data, the values are discrete and ordinal, and the scale has specific meaning and interpretation. Each value on the scale corresponds to a distinct response from the participants, and there is no continuous underlying measurement. The concept of outliers is more applicable to continuous variables where extreme values could be considered as potential outliers. In the case of Likert scale data, it is essential to preserve the integrity of the scale and respect the ordinal nature of the responses.

Four task time variables in which task time was recorded were considered ratio variables. In the context of time, a ratio variable represents a measurement where zero represents the absence of the quantity being measured (e.g., zero seconds means no time has elapsed), and the intervals between different time points are consistent and meaningful. Basic descriptive statistical analysis technique was used to analyze time in which average mean time rating for each task individually and then overall for all task times was computed and plotted as bar graph.

For testing the hypothesis 95 percent confidence level was considered. For graphical representation, bar plot was used to compare mean ratings. Also, for Attrakdiff questions the likert scale values were from -3 to +3 for which when p-value was computed using same rating there was a warning displaying "Exact p-value calculation does not work if there are zeros and negative values." so to cater for it a function was written using which values were converted from -3 to +3 to 1 to 7 scale and then p-value was computed.

5 Results

5.1 RQ1 - Spatial Presence

The resulting p-value after analysis came out to be p-value: 0.05 which proved the hypothesis insignificant. The graphical representation of the result is in Fig. 9.

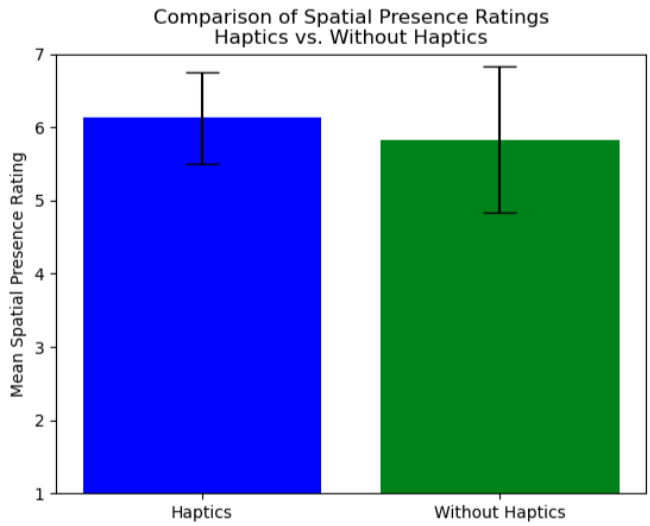

Fig. 9 Mean Spatial Presence rating comparison for haptics and non-haptics environments.

5.2 RQ2 - Perceptual Realism

The resulting p-value after analysis came out to be p-value: 0.11 which proved the hypothesis insignificant. The graphical representation of the result is in Fig. 10.

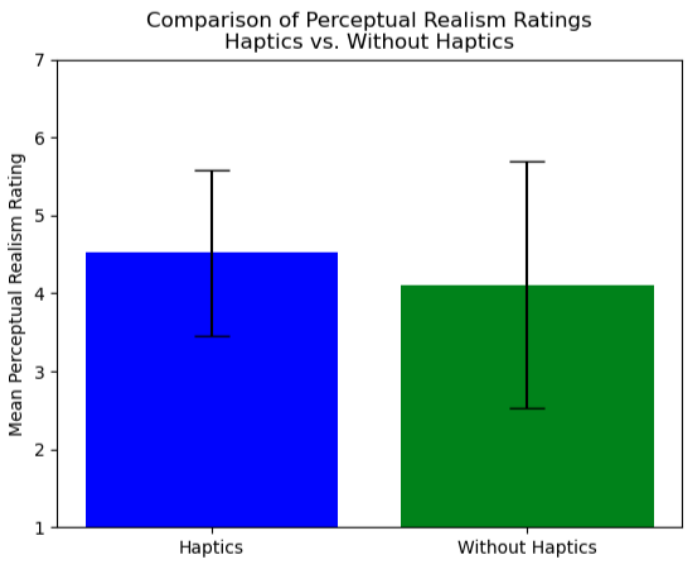

Fig. 10 Mean Perceptual Realism rating comparison for haptics and non-haptics environments.

5.3 RQ3 - Mental Immersion

The resulting p-value after analysis came out to be p-value: 0.09 which proved the hypothesis insignificant. The graphical representation of the result is in Fig. 11.

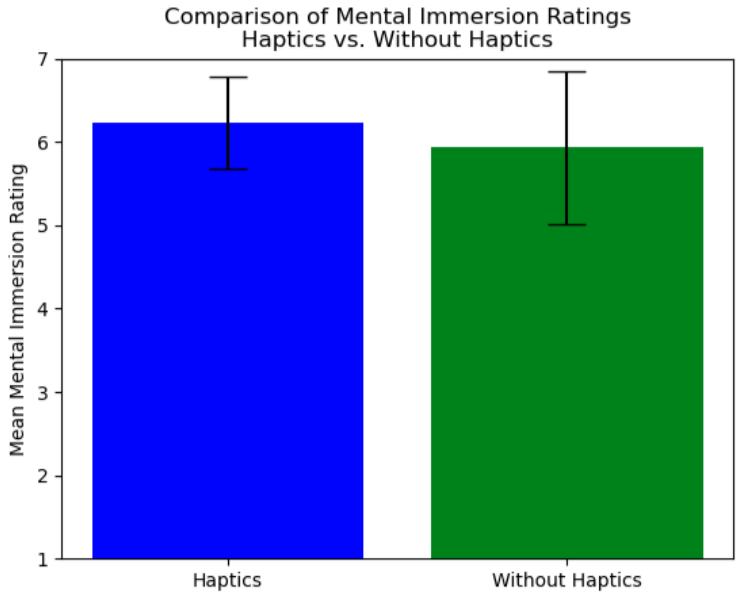

Fig. 11 Mean Mental Immersion rating comparison for haptics and non-haptics environments.

5.4 RQ4 - User Experience

For user experience data is analyzed for each variable, comparing mean, and shown in bar graph (Fig. 12). Then data is grouped into three groups as per Attrakdiff standard, columns 1 to 4 grouped as pragmatic quality, column 5-8 as hedonic quality and last 2 as overall attractiveness and displayed on the graph (Fig. 13). The resulting p values for these groups were p_value_pragmatic: 0.25, p_value_hedonic 0.46, p_value_attractiveness: 0.17 which proved the hypothesis insignificant.

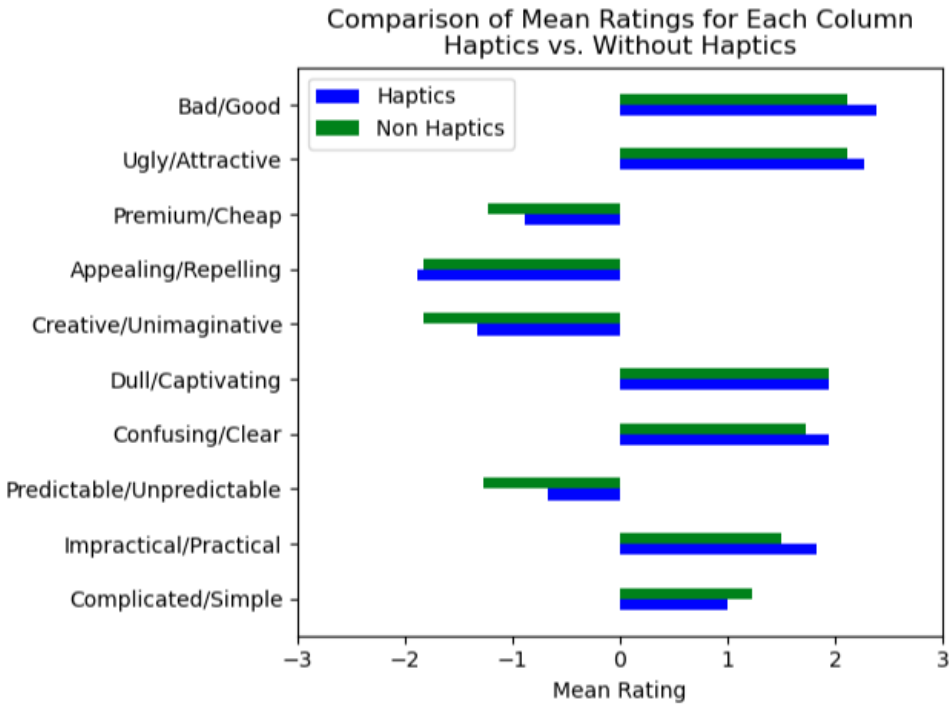

Fig. 12 User Experience rating comparison for haptics and non-haptics environments.

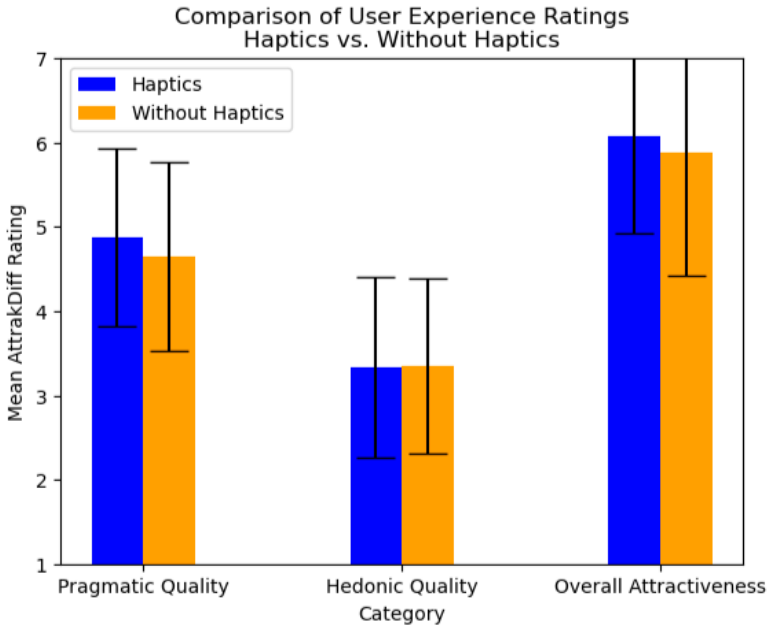

Fig. 13 User Experience based on three grouped Attrakdiff categories.

Attributes	Mean Haptics	Mean Non-Haptics	SD Haptics	SD Non-Haptics
Bad/Good	2.38	2.11	0.77	1.07
Ugly/Attractive	2.27	2.11	0.82	1.23
Premium/Cheap	-0.88	-1.22	2.02	1.66
Appealing/Repelling	-1.88	-1.83	1.56	1.33
Creative/Unimaginative	-1.33	-1.83	2.19	1.38
Dull/Captivating	1.94	1.94	0.99	1.16
Confusing/Clear	1.94	1.72	1.10	1.27
Predictable/Unpredictable	-0.66	-1.27	1.45	1.56
Impractical/Practical	1.83	1.50	1.24	1.50
Complicated/Simple	1.00	1.22	1.49	1.47

Table 3. Mean and Standard Deviation Values for User Experience

5.5 RQ5 - Task Performance

The resulting average mean time for task 1 came out to be almost the same. For task 2 the average mean time taken in haptics was more than non haptics. Task 3 and task 4 average mean time taken in haptics was less than non haptics. Overall meantime comparison shows that participants have taken less time for tasks with haptics experience than with non-haptics.

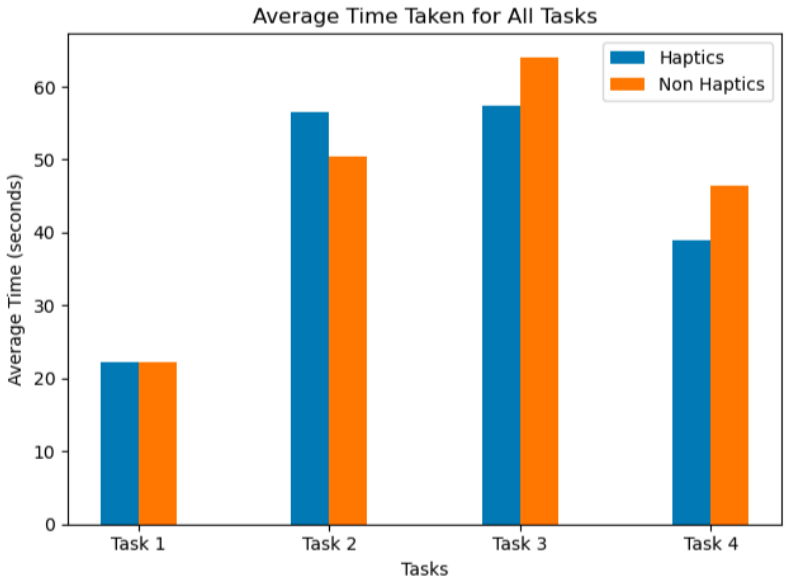

Fig. 14 Average mean time comparison for each task for haptics and non-haptics

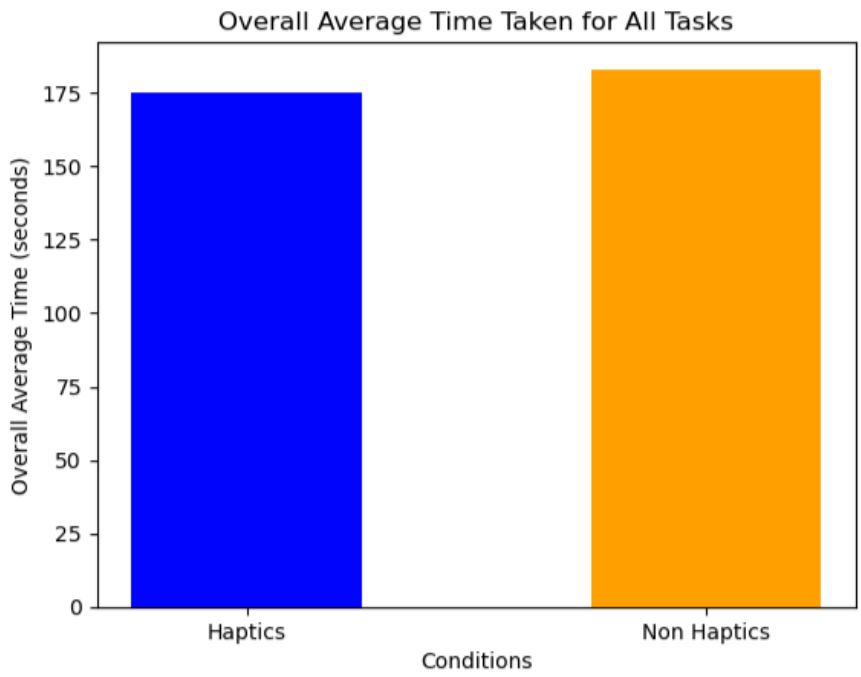

Fig. 15 Overall average mean time comparison of all four tasks time collectively.

5.6 Summarized Results

Dependent Variables	Mean Haptics	Mean Non-Haptics	SD Haptics	SD Non-Haptics	p-value
Spatial Presence	6.12	5.83	0.62	0.99	0.05
Perceptual Realism	4.52	4.11	1.06	1.58	0.11
Mental Immersion	6.23	5.93	0.55	0.91	0.09
User Experience (Pragmatic Quality)	4.87	4.65	1.04	1.12	0.25
User Experience (Hedonic Quality)	3.33	3.34	1.06	1.03	0.46
User Experience (Attractiveness)	6.08	5.88	1.16	1.47	0.17

Table 4. Summarized results of variables

Task No.	Mean Average Time Haptics	Mean Average Time Non-Haptics
Task 1	22.27	22.22
Task 2	56.50	50.38
Task 3	57.33	64.05
Task 4	38.88	46.44

Table 5. Average mean time values for each task in haptics and non-haptics condition

6 Discussion

For testing the hypothesis considering our first research question focused on spatial presence, the resulting p-value came out 0.05, the hypothesis “With the haptic feedback users will report higher levels of spatial presence within the environment compared to when there will be no haptic feedback.” proved insignificant. Although it was observed that the value of p: 0.05 was quite close and the hypothesis got rejected with very slight margin, the overall mean values for haptics rating were high and even it is evident in the graphical representation (Fig. 9) as well, that people have rated spatial presence better for haptics as they have the haptic feedback and felt more present in the Virtual environment. Also, the mean and standard deviation values show in the results that haptics environment was rated better (Table 4). Our findings align with the previous studies, such as the research titled “Evaluation of Different Types of Haptic Feedback Influencing the Task-based Presence and Performance in Virtual Reality”. The study aimed to examine the influence of different types of haptic feedback on presence and performance regarding manual tasks in VR. A comprehensive user study involving 14 subjects, who performed throwing, stacking and object identification tasks in VR with visual (i.e., sensory substitution), vibrotactile, or force feedback. The degree of presence and task-related performance metrics were measured. The same HTC Vive and Sense Gloves were used for the experiment execution. The result they reported is “force feedback did not significantly increase the presence (factors) or reached the maximum scores, though the box plots labeled with THR suggest this. Therefore, we need to reject our corresponding hypothesis at this point. We assume that this became not not yet significant due the rather small number of subjects and hardware issues with the force exoskeleton, which impeded the subjects’ hand motion and needed to be occasionally re-started. The same character applies to vibrotactile feedback, but here the limited "eld-of-view of the LMC can be mentioned as main reason”[7].

For perceptual realism, the resulting p-value came out as 0.11 which proved the hypothesis “Users who experience the haptic feedback will evoke higher perceptual realism, compared to the absence of it.” insignificant. A possible reason for this resulting p-value is that visual for both environments was almost the same and that’s why users have perceived the reality of objects in both environments almost the same, but the overall mean values for haptics rating were high and even it is evident in the graphical representation (Fig. 10) as well that people have rated perceptual realism better for haptics and perceived objects to be more realistic than for the haptics.

For mental immersion, the resulting p-value was 0.09 which doesn’t fulfill the confidence level requirement and hence the hypothesis “Users who experience haptic feedback will feel more mentally immersed in the environment compared to when there will be no haptic feedback.” proves as in-

significant. A possible reason for this resulting p-value is that for both environments the tasks were same and user engagement within the environment to feel mentally immersed was almost the same as they performed same activity and that is why users have felt almost the same mental immersion in both environments, but the overall mean values for haptics mental immersion rating were high(Table 4) and even it is evident in the graphical representation as well that people have rated mental immersion better for haptic feedback environment.

For user experience considering graphical results (Fig. 12) looking at individual responses, it was observed that users have rated the haptics environment better than the non-haptics for some variables and even its reverse that they have rated non-haptics better too for some of the qualities. They have found the haptics environment a bit more attractive than the non-haptics, also they have experienced haptics product more appealing and more practical. But at the same instance some of the ratings were higher for non-haptics such as participants felt that the non-haptics environment was more simple, more predictable, more premium, and possible reason we found that as things were grabbing and breaking easily, they found it more easier. From Fig. 13 considering the grouped Attrakdiff quality in which for pragmatic qualities and overall attractiveness of the product, there is no significant difference between haptics and non-haptics but overall, the mean rating for haptics is a bit higher. Our findings align with previous studies, such as the research titled "Expectations for user experience in haptic communication with mobile devices." This study aimed to understand users' expectations concerning their experience with haptic devices and applications, particularly in the context of person-to-person communication via mobile devices using haptic feedback. The focus of this research was to gain valuable insights into how haptic communication influences user experiences. The results stated as “All in all participants believed that using the sense of touch can bring something new into the interaction with mobile devices. Touch was regarded as very natural as an interaction modality. Here, haptic interaction included both output and input modalities. Despite the quite positive expectations, participants expected they would not use haptics very often”[10]. This shows that even in the case of other studies people have preferred non haptics environments.

Considering task performance from the results (Fig. 14) in the form of mean and standard deviation values the behavior seen is that for task 1 participants have taken almost the same time in haptics and non haptics environments. For task 2 the time for haptics is a bit high and on observing data manually it is concluded that most of the participants who performed haptics first were confused as they did not even know how the gardening tools such as shovel, pitchfork and rake look like. They read the task but were still confused with the names of what exactly that thing is, so they have taken more time. For task 3 less time was taken for haptics environment compared to non haptic environment and same goes for task 4 where participants took less time in haptics feedback. On comparing the

overall average mean time ratings (Table 3) for haptics and non-haptics tasks performances we figured out that, for haptics environment participants were quick as compared to non-haptics which proves that haptic feedback has helped the users with task performance and hence hypothesis is proven significant. While searching for other conducted studies where task performance time is monitored, a study titled as “Effect of haptic feedback in laparoscopic surgery skill acquisition” in which the user has to perform surgical tasks in 2 conditions first with haptics and then without haptics. Results reported as “When the learning rate between the haptics and non-haptics groups were compared, the results from the one-way ANOVA showed no significant differences overall ($F[1, 18] = 4.02$; $p < 0.061$). The slightly higher learning rate in the haptics group (70% vs 64% in the no-haptics group) indicates that these subjects experienced a flatter learning curve”[11].

7 Conclusions and Future Work

The study was conducted using 18 participants to investigate the effect of vibrotactile and force feedback on spatial presence, perceptual realism, mental immersion, user experience and task performance. The result we had after the analysis didn't show any significant differences for all 5 metrics, however the overall graphical representation shows that the mean ratings for haptics environments are higher than for non-haptics. We summarized the results in table 4 and table 5 for all the metrics and comparing the mean and standard deviation values found the haptics a bit better than non-haptics overall. Even compared to other past studies it is concluded that for users, the haptic feedback is not always desirable, many times users have rated non-haptics better and highlighted the preference of using non-haptics due to the ease of use and simpler task performance.

For our study, after completing the whole experiment, participants were asked verbally which product they liked more, and which one was easy to use for them. Most of the participants reported that they liked the haptic feedback environment more as it feels more realistic and practical but at the same time some participants reported that for them the haptic environment was more complicated. They also said that non-haptic was more simple to use and predictable for them.

Considering the task performance time, it was a bit better for haptics than for non-haptics environment although there was not a significant difference between the time ratios, somewhat similar trend seen in other studies as well where users reported that over longer time span compared to haptics, the non-haptics experience is better and easy for them as it requires less energy and mental engagement.

Another possible reason for the not so significant difference is that our data set was small involving just eighteen people, there is the possibility that in the future if the same experiment will be performed for a greater number of participants some significant difference can be computed from results.

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List of Abbreviations

HMD	Head Mounted Display
VR	Virtual Reality
CSV	Comma separated Values
Fig.	Figure
TPI	Telepresence Inventory questionnaire
SPES	Social Presence Experience Scale

Declaration of Independence

We Declare that we have carried out and written this work independently. Sources, literature and tools used by us are marked as such.

Signatures

Qasim Saboor

Hamd Mehfooz Khan

Distribution of Work

Team Member	Contribution
Qasim Saboor, Hamd Mehfooz Khan	Research and implementation approach
Qasim Saboor, Hamd Mehfooz Khan	Environment Design
Qasim Saboor, Hamd Mehfooz Khan	Implementation of breakable and deformable objects
Qasim Saboor, Hamd Mehfooz Khan	Implementation of Grabbable and throwable objects
Qasim Saboor	Implementation of the particle system
Hamd Mehfooz Khan	Character animation with sound
Hamd Mehfooz Khan	Plant growth in stages
Qasim Saboor, Hamd Mehfooz Khan	Implementation of Auditory cues on trigger events
Hamd Mehfooz Khan	Finger Specific Feedback
Qasim Saboor	Implemented Timer and navigation arrows
Hamd Mehfooz Khan, Qasim Saboor	Added Vibrotactile and force feedback on the objects and trigger events
Qasim Saboor, Hamd Mehfooz Khan	Implementation of Arm Swinging Locomotion
Qasim Saboor, Hamd Mehfooz Khan	Implementation of AI based NavMesh
Hamd Mehfooz Khan, Qasim Saboor	Implementation of the tasks as per study design
Qasim Saboor, Hamd Mehfooz Khan	Bug Fixing
Qasim Saboor, Hamd Mehfooz Khan	Testing
Hamd Mehfooz Khan	Experimental Study Design
Hamd Mehfooz Khan	Questionnaire Design
Qasim Saboor, Hamd Mehfooz Khan	Planned Study, Conducted Study
Qasim Saboor, Hamd Mehfooz Khan	Data Analysis
Qasim Saboor	Report - Introduction, Related Work, Implementation, Conclusion, Future Work
Hamd Mehfooz Khan	Report - Implementation, Evaluation, Result, Discussion